

I didn't proofread carefully. Please use with caution.

----- ★ -----

1. 2D rotations.

In the previous problem set we found that rotations of the coordinate frame around the z axis are described by the matrices

$$R_z(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

If we restrict to rotations around the z axis, we can think of these as 2D rotation matrices. This is a one-parameter family of matrices, parametrized by the rotation angle θ .

- (a) Is multiplication of these 2D rotation matrices commutative? Demonstrate using matrix multiplication, and then explain physically why. (If needed, look up what 'commutative' means.)

(Partial) Solution/Hint →

Commutative means the order of multiplication does not matter. We have already found out (previous problem set) that $R_z(\theta)R_z(\phi) = R_z(\theta + \phi)$; so

$$R_z(\theta)R_z(\phi) = R_z(\theta + \phi) = R_z(\phi + \theta) = R_z(\phi)R_z(\theta)$$

Hence the 2D rotation matrices are commutative.

Physically, if you are rotating around the same axis, rotating first by angle θ and then by angle ϕ is the same as rotating first by angle ϕ and then by angle θ .

— * —

- (b) Look up and report the definition of a GROUP in mathematics (basic abstract algebra).

(Partial) Solution/Hint $\rightarrow$

A set G is a group under the binary operation $\star$ if and only if the following conditions hold.

- **Closure.** If a and b are elements of G , then so is $a \star b$.
- **Associativity.** For all a, b and c in G ,

$$(a \star b) \star c = a \star (b \star c)$$

- **Identity element.** There exists an element e such that

$$e \star a = a \star e = a$$

for every element of a .

- **Existence of inverse.** For every element a of G , there exists an inverse element d , such that

$$d \star a = a \star d = e$$

— —

Groups are fundamental to physics and to mathematics. The definition above with four properties might appear terribly abstract. However, examples of this structure appear in many places in mathematics and physics. For example, the set of all integers ($\mathbb{Z}$) is a group under addition, but not under multiplication. The set of positive integers (set of natural numbers, $\mathbb{N}$) is not even a group under addition. Please do verify these statements.

In physics, groups usually appear for transformations or for operations such as symmetry operations. Since transformations can often be represented as matrices, we are particularly interested in groups of matrices under matrix multiplication.

Matrix multiplication is associative, so this property is automatically satisfied for any set of matrices. To check whether a set of matrices is a group under matrix multiplication, one has to check the other three group properties.

— * —

- (c) Show that the 2D rotation matrices form a group under multiplication. Make sure you explicitly demonstrate all aspects of the definition of a group (closure, associativity, identity and inverse.)

(Partial) Solution/Hint →

We have to check for all four properties that make a group.

- **Closure.** We have already seen that two 2D rotation matrices, multiplied, give another 2D rotation matrix:

$$R_z(\theta)R_z(\phi) = R_z(\theta + \phi)$$

Thus closure is satisfied.

- **Associativity.** Matrix multiplication is associative.
- **Identity element.** The 2×2 unit matrix is part of our set, because it represents rotation by angle zero, $I = R_z(0)$. This serves as the identity element, since

$$R_z(\theta)R_z(0) = R_z(0)R_z(\theta) = R_z(\theta)$$

for any rotation matrix $R_z(\theta)$.

- **Existence of inverse.** For any rotation matrix $R_z(\theta)$, the matrix $R_z(-\theta)$ representing the opposite rotation is its inverse, because

$$R_z(\theta)R_z(-\theta) = R_z(-\theta)R_z(\theta) = I$$

— —

All the group properties are satisfied, so the set of 2D rotation matrices under matrix multiplication is a group.

Incidentally, the group of 2D rotations is known as $SO(2)$. It is the set of all orthogonal matrices which have determinant $+1$.

If reflections are included, we get a larger group which is called $O(2)$. This is the set of all orthogonal matrices, with determinant ± 1 .

Note that the number of elements in $SO(2)$ is infinite. This is an infinite group. The groups we will meet in Special Relativity are all infinite groups. (They are examples of Lie groups: feel free to look up what that means.) Finite groups are also common in physics. Can you construct an example of a finite group?

— * —

- (d) Explain whether the group of 2D rotations is abelian or non-abelian. (If necessary, look up what these mean.)

(Partial) Solution/Hint $\rightarrow$

A group is called abelian or commutative if the group operation is commutative for any two members of the group. Since we have shown that

$$R_z(\theta)R_z(\phi) = R_z(\phi)R_z(\theta)$$

i.e., that multiplication of 2D rotation matrices is commutative, the 2D rotation group is abelian.

Note that commutativity is NOT required for a set to be a group. For example, 3D rotations form a group, but is not an abelian group.

Please also note that you are supposed to know and remember what ‘associative’ and ‘commutative’ mean, without confusing the two:

$$\text{Associative:} \quad A * (B * C) = (A * B) * C$$

$$\text{Commutative:} \quad A * B = B * A$$

They are utterly different properties.

$$__ * __$$

- (e) We have found that, under 2D rotation of the coordinate axes, the components of the displacement $\mathbf{r}$ transform via the matrix given above. Argue why the components of any vector (e.g., velocity, force, angular momentum,...) would transform the same way under the same rotation.

(Partial) Solution/Hint $\rightarrow$

The rotation transformation was derived based on the fact that, if the displacement $\mathbf{r}$ points at angle ϕ with respect to some axis, it has components $r \cos \phi$ and $r \sin \phi$ parallel and perpendicular to that axis. Now any vector $\mathbf{A}$ (be it a velocity, force,...) has components $A \cos \phi$ and $A \sin \phi$ in directions parallel and perpendicular to an axis with respect to which the direction of $\mathbf{A}$ is at angle ϕ . Hence the components of $\mathbf{A}$ should transform exactly the same way as the components of $\mathbf{r}$.

In other words, if the x' - y' coordinate system is obtained by rotating the x - y system by angle θ , and if

$$\mathbf{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} = A'_x \hat{i}' + A'_y \hat{j}' + A'_z \hat{k}'$$

then

$$\begin{pmatrix} A'_x \\ A'_y \\ A'_z \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$$

In fact, this may be thought of as the *definition* of a vector: a vector is a three-component object such that the three components transform the same way as the components of displacement vectors.

— * —

2. [6 pts.] What is the determinant of an orthogonal matrix?

What do positive and negative determinant physically mean for rotation matrices?

Hint: First find out (or recall) whether the determinant of the product of two matrices is the product of their determinants.

(Partial) Solution/Hint →

The determinant of the product of two matrices is the product of their determinants. The determinant of the transpose of a matrix is equal to the determinant of the matrix. The determinant of the unit matrix is +1.

Using these facts, the defining feature of orthogonal matrices

$$A^T A = I$$

yields

$$1 = \det(I) = \det(A^T A) = \det(A^T) \det(A) = \det(A) \det(A) = \det(A)^2$$

so that

$$\det(A) = \pm 1$$

Thus orthogonal matrices have determinant +1 or -1. This is important: it may even be worth remembering.

Rotation matrices have to be orthogonal because otherwise the vector magnitude is not preserved. Hence rotation matrices have determinant +1 or -1. These represent respectively *proper* rotations and *improper* rotations. Improper rotations are those involving not only a rotation but also a *reflection*. Under an improper rotation, if we start with a right-handed coordinate system ($\hat{i} \times \hat{j} = \hat{k}$), we end up with a left-handed coordinate system ($\hat{i}' \times \hat{j}' = -\hat{k}'$). Generally, considering this type of transformation is not helpful in the study of physical rotations, so one often restricts to the set of proper rotations.

— * —

3. Rotations in 3D.

- (a) By analogy, write the transformation matrix for counterclockwise rotation around the x -axis (rotation of the y - z plane). In other words, write out the 3×3 matrix $R_x(\phi)$.

(Partial) Solution/Hint $\rightarrow$

Noting that

$$R_z(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

has only a diagonal 1 in the z rows and columns, and otherwise has the 2×2 matrix of trigonometric functions in the x - y block, we infer that $R_x(\phi)$ will have only a diagonal 1 in the x rows and columns, and otherwise will have a 2×2 matrix of trigonometric functions in the y - z block:

$$R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{pmatrix}$$

___ * ___

- (b) Show that $R_z(\theta)$ and $R_x(\phi)$ do not commute.

(Partial) Solution/Hint $\rightarrow$

Calculate $R_z(\theta)R_x(\phi)$ and $R_x(\phi)R_z(\theta)$; show that you get different matrices. I will not type this out.

___ * ___

- (c) The set of all possible 3D rotations (by any angle, around any axis) is known to be a group. Explain whether this group is abelian or non-abelian.

(Partial) Solution/Hint →

We have found in the previous problem that two rotation operations (rotation matrices) do not commute when the two rotations are not around the same axis.

Hence, not every pair of members of the 3D rotation group commute. This means that the 3D rotation group is non-abelian.

— * —

- (d) An arbitrary 3D rotation matrix can be constructed by a combination of rotations around the x -, y - and z - axes. Using this fact, argue that all 3D rotation matrices are orthogonal. Using the same fact, argue that all rotation matrices have unit determinant.

Hint: First find out whether the product of two orthogonal matrices also orthogonal.

(Partial) Solution/Hint →

The product of two orthogonal matrices is orthogonal: if $A^T A = I$ and $B^T B = I$, then

$$\begin{aligned}
 (AB)^T(AB) &= (B^T A^T)(AB) \\
 &= B^T(A^T A)B \quad \left\{ \begin{array}{l} \text{because matrix} \\ \text{multiplication} \\ \text{is associative} \end{array} \right. \\
 &= B^T I B = B^T B \\
 &= I
 \end{aligned}$$

This can easily be extended to state that the product of three orthogonal matrices is orthogonal. Since 3D rotation matrices can be expressed as the product of three 2D rotation matrices, which we have found previously to be orthogonal, this means that all 3D rotation matrices are orthogonal.

— * —

- (e) If $\mathbf{A}$ and $\mathbf{B}$ are two vectors, show that the scalar product $\mathbf{A} \cdot \mathbf{B}$ is invariant under rotation of coordinate frames, Use the fact that the rotation matrix R is orthogonal.

(Partial) Solution/Hint $\rightarrow$

Since all vectors transform the same way as position vectors, we have

$$\begin{pmatrix} A'_x \\ A'_y \\ A'_z \end{pmatrix} = R \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \quad \text{and} \quad (A'_x \ A'_y \ A'_z) = (A_x \ A_y \ A_z) R^T$$

and the same relations for the transformation of the components of $\mathbf{B}$.
Hence

$$\begin{aligned} \mathbf{A}' \cdot \mathbf{B}' &= (A'_x \ A'_y \ A'_z) \begin{pmatrix} B'_x \\ B'_y \\ B'_z \end{pmatrix} \\ &= (A_x \ A_y \ A_z) R^T \times R \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \\ &= (A_x \ A_y \ A_z) I \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \\ &= (A_x \ A_y \ A_z) \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \mathbf{A} \cdot \mathbf{B} \end{aligned}$$

— * —

- (f) We've been using rotation matrices to describe the rotation of coordinate axes. Is this an *active transformation* or a *passive transformation*? Explain why.

(Partial) Solution/Hint →

Active transformations are those where an object, or physical points, are moved or rotated. Passive transformations are those where the coordinate system is changed (moved or rotated), but the physical system remains the same. In other words, only the description changes in passive transformations, not the physical points or any physical object. The descriptions of the two types of transformations are very similar. In these two problem sets, we have been using two different coordinate systems to describe the same physical space. These were passive transformations.

— * —